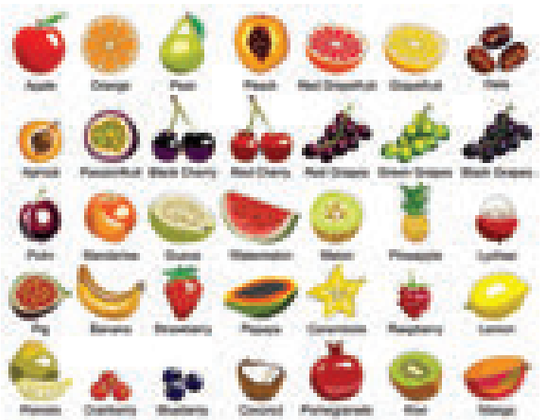
Water – soluble vitamins

Water – soluble vitamins are essential of health, and each one has its own function in the body and due to its solubility in water. It is normally easily lost in urine. The water soluble vitamins include vitamin C and 8 B complex vitamins – thiamine, riboflavin, niacin, pyridoxine, folate, cyanocobalamin, biotin and pantothenic acid. Most of these are unstable and thus easily destroyed by exposure to water, heat, air or light.

Vitamin C

Many fruits and vegetables are rich in vitamin C, also known as ascorbic acid. To ensure optimal physiological functioning and to prevent subclinical deficiencies, a person needs to consume a good source of vitamin C every day because the body normally stores only small amounts of Vitamin C.



Sources of Vitamin C

Functions

- Vitamin C is required in the synthesis of collagen in connective tissue, steroid hormones, carnitine, etc.
- Vitamin C is required for the conversion of cholesterol to bile acids.
- It enhances iron bioavailability.
- Ascorbic acid is a great antioxidant and helps to protect the body against pollutants.
- Ascorbic acid also promotes healthy cell development.
- Vitamin C is essential for the formation and maintenance of intercellular cement substances such as bone matrix, cartilage dentine, collagen, connective tissue, etc.
- Vitamin C is needed for healthy gums and to protect against infection.

Food sources

Good sources of vitamin C are green leafy vegetables, berries, citrus fruits, guavas, tomatoes, melons, papayas, etc.

Deficiency

In infants and children, vitamin C deficiency results in defective bone formation leading to retardation of growth.

B – complex vitamins

The eight B – complex vitamins include thiamine (B1), riboflavin (B2), niacin (B3), pyridoxine (B6), cyanocobalamin (B12), folic acid, biotin (B7), and pantothenic acid. The B vitamins are easily lost in cooking water because they are water soluble. With the exception of niacin, all other B vitamins lose some activity when exposed to heat, oxygen, light, or alkaline conditions.

Thiamine (Vitamin B1)

Thiamine also called B1 is used in many different body functions and deficiencies may have far reaching effects on the body, yet very little of this vitamin is stored in the body and depletion of this vitamin can happen within 14 days. Thiamine is integrally involved as a coenzyme in the catabolism of carbohydrates to yield energy.

Functions

- It is also required for the health of the nervous system.
- It is used in the manufacture of hydrochloric acid, and therefore plays a part in digestion.

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